Ria Banerjee

Full Stack Software Developer
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PROFESSIONAL SUMMARY

Full Stack Developer with expertise in Python, Go, Java, and JavaScript (React, Node). Expertise in designing test-driven RESTful APIs with seamless database integration and optimal performance. Experienced in cloud architecture, containerization, and CI/CD pipelines for efficient deployments. Adept at designing real-time data processing systems using pub/sub architectures like Kafka and MQTT. Integrated over 20 third-party tools to enhance scalability and streamline system functionalities. Collaborative, solution-oriented, seeking to build innovative system solutions.

TECHNICAL SKILLS

- *Programming Languages*: Python, Go, Java, C# .Net, C++, C, Shell/Bash.
- *Web Development*: TypeScript, React, Node.js, HTML5, CSS, ElectronJS.
- *Frameworks & Tools*: Django, Flask, SpringBoot, Docker, Kubernetes, RESTful API, Kafka, MQTT, RabbitMQ, AWS, GCP, Git, Jenkins, Ansible, ServiceNow, Twilio, OAuth, ReCAPTCHA.
- *Database Technologies*: MySQL, MongoDB, PostgreSQL, SQLAlchemy, GraphQL, Redis, DynamoDB.
- *Cloud & DevOps*: GCP, AWS(EC2, S3, ECR+ECS), Docker, Kubernetes clusters, Jenkins, CI/CD, Nexus Lifecycle, GitOps.
- *Security and Monitoring*: Grafana, Kibana, Vault, SSL/TLS configuration.
- *Machine Learning*: Pandas, TensorFlow, OpenAI APIs, Data Modeling, PyTorch, Scikit-learn.

WORK EXPERIENCE

Full Stack Developer, EAGL Technology Inc. *May 2024-present | Albuquerque, NM*
Tech Stack: Python, Go, React, SQL, Redis, Docker, Flask, Kafka, MQTT
Project Description: *A gunshot detection system with sensors deployed in critical areas like schools and hospitals. The system enables users to receive real-time gunshot alerts and analysis. Currently transitioning from on-premises to a cloud-based solution, including server setup, environment configuration, caching, and load balancing.*

- Led a complete overhaul of legacy gunshot detection software into a microservice architecture, using Flask-based REST APIs, deployed on Kubernetes + Cloud, enhancing system efficiency by 80%.
- Architected API Gateways for optimized communication channels using publisher-subscriber models (MQTT and Kafka), task queues, setting Rate Limit, reverse proxy (Nginx), and improving event processing speed.
- Implemented E911 service using Twilio, set asynchronous message queues using Celery, implemented cache with Redis, ORM using SQLAlchemy, and managed MySQL DB on DigitalOcean.
- Built a CI/CD pipeline using Jenkins, Ansible, and Docker, automating deployments, accelerating release cycles by 60%.
- Configured and managed SSL/TLS certificates to secure client-server communications.

Full Stack Developer, BabelCast Inc. *Oct 2023-Jan 2024 | Chicago, USA (Remote)*
Tech Stack: Python, Flask, React, Node, AWS, REST, Docker, Kubernetes
Project Description: *Worked as a founding engineer for an entertainment-gaming startup, enabling users to transpose their voice into thousands of voice models, including celebrities and custom models. The app supported real-time voice conversion during calls and allowed users to create, share, and collaborate on voice models.*

- Led the development of Python and Flask backend, scaling to handle **100K+ concurrent users**, deployed using Docker and Kubernetes clusters on AWS.
- Incorporated over 20,000 voice models, integrated with GraphQL schema, enhancing database interactions and reducing query response time by 50%. Utilized Redis for in-memory cache, and AWS S3 for blob storage.
- Led OAuth integration for secure access and seamless model deployment.
- Orchestrated testing frameworks (load, stress, end-to-end tests) using Gatling, Locust, and custom SocketIO cases.
- Collaborated with UI/UX for front-end and back-end integration.

Graduate Research Assistant, UNC Charlotte *Mar 2023-Oct 2023 | Charlotte, NC*
Tech Stack: Python, SQL, React, TensorFlow, Machine Learning models
Project Description: *Collaborated with a healthcare research group to analyze biomedical data. The goal was to develop a machine learning model capable of predicting anxiety under different conditions, with applications aimed at supporting individuals with disabilities.*

- Created ML backend in Python to analyze physiological data, using Logistic Regression and K-Nearest Clustering **achieving 80% accuracy.**
- Developed React React-based dashboard, supporting visualizations of data, and supporting manipulation of current and historical data.

Systems Engineer, Equifax Inc. via Tata Consultancy Services *May 2021-Aug 2022 | Remote, India*
Tech Stack: Java, SpringBoot, Maven, GCP, Kubernetes, SQL, Grafana
Project Description: *The project involved optimizing on-premise infrastructure and leading the cloud migration, including traffic redirection and safe decommissioning of legacy systems.*

- Engineered microservices for the Equifax UK system using SpringBoot, enhancing the UI and application features, resulting in a 150% boost in user satisfaction scores.
- Part of the migration team for legacy systems to GCP migration, architecting Kubernetes clusters and configuring Grafana for monitoring, achieving 200% improvement in uptime and reliability.
- Implemented CI/CD practices using Jenkins and ServiceNow for microservice deployments, achieving a 95% release success.
- Conducted code quality checks and vulnerability management using Nexus Lifecycle and SonarQube.

PROJECTS

Personal Portfolio Project
Created a personal portfolio website using React+Node+Vite+Nginx (frontend) and Python+Flask (backend). Utilized Google ReCAPTCHA, integrated Rate Limiting, SSL certs, and reverse proxy. Dockerized and deployed to AWS ECR + EC2.
Large Language Model (LLM) based Chatbot in Python
Created a Chatbot trained on personal resume using Retrieval-Augmented Generation (RAG) integrated with Django-based web-app, creating REST endpoints, with React, hosted on DigitalOcean.

PUBLICATION

Recommendation Systems based on Collaborative Filtering using Autoencoders: Issues and Opportunities, published in Recent Innovations in Computing, Springer. DOI: https://doi.org/10.1007/978-981-15-8297-4_32.

EDUCATION

University of North Carolina at Charlotte, USA | *Master's in Computer Science* Aug 2022- May 2024
Nirma University, India | *Master's of Computer Application* Aug 2019- May 2021
Ahmedabad University, India | *Bachelor of Computer Application* Aug 2016- May 2019